

Professor: Explain your project.

You:

"My project is called **ThreatShield**. It is an AI-based Web Attack Detection System. It uses a Deep Learning model (Bi-Directional LSTM) to analyze HTTP requests and identify whether they are normal or malicious. The system can detect five classes:

- Normal
 - SQL Injection
 - Cross Site Scripting (XSS)
 - Path Traversal
 - Phishing URL"
-

Professor: Show me how it works.

Step 1 — User enters a request

For example:

```
<script>alert('Hello')</script>
```

or

```
SELECT * FROM users WHERE username='admin'--
```

The user pastes it into the text box.

Then clicks

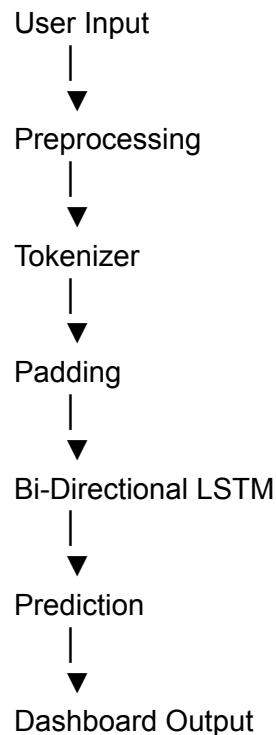
Analyze Request

Professor: What happens after clicking Analyze?

You can explain:

"As soon as I click Analyze, the request is sent to the Deep Learning model."

Flow:



Professor: What is preprocessing?

Answer:

"The request is converted into lowercase if needed, cleaned, then the tokenizer converts every word into numbers because a neural network cannot understand text directly."

Example

Input

`<script>alert(1)</script>`

becomes

[45, 302, 98, 17, ...]

Professor: Why convert into numbers?

"Neural networks only understand numerical values. Therefore, the tokenizer converts text into integer sequences."

Professor: What is padding?

Answer:

"Different requests have different lengths. The model expects every input to have the same size. Padding adds zeros to shorter sequences so every input has the same length."

Example

[34,18,55]

becomes

[34,18,55,0,0,0,0,0...]

Professor: What does the Deep Learning model do?

"The padded sequence is passed to the Bi-Directional LSTM model. The model has already learned patterns from thousands of training samples and predicts which attack category the request belongs to."

Professor: What output does the model produce?

The model gives probabilities.

Example

Attack	Probability
Normal	1%
SQL Injection	2%
XSS	95%
Path Traversal	1%
Phishing	1%

Then the highest probability is selected.

Result:

Attack:
Cross Site Scripting

Confidence:
95%

Professor: After prediction what happens?

Now explain every output.

1. Attack Type

Shows

Cross Site Scripting

Meaning

"The model believes this request is an XSS attack."

2. Confidence

Example

95%

Meaning

"The model is 95% confident about its prediction."

3. Prediction Time

Example

22 ms

Meaning

"The system only took 22 milliseconds to analyze the request."

4. Probability Graph

Explain:

"This graph shows the probability assigned to every attack class by the neural network. The highest bar is the predicted attack."

Example

Normal 1%

SQL 2%

XSS 95%

Traversal 1%

Phishing 1%

5. AI Attack Explanation

Example

Bi-Directional LSTM detected
JavaScript execution patterns.

Explain:

"This is a human-readable explanation generated from the model's prediction so
users understand why the request was classified this way."

6. Threat Severity

Shows

HIGH

Explain

"Every attack category is mapped to a severity level."

Example

Normal

↓

LOW

SQL Injection

↓

CRITICAL

XSS

↓

HIGH

7. Threat Score

Example

92%

Explain

"Threat Score combines the attack severity and the model's confidence to estimate the overall risk."

8. Adaptive Response

Example

BLOCK

or

RATE LIMIT

or

MONITOR

Explain

"Based on severity, the system recommends what action should be taken."

9. Logging

Explain

"Every analyzed request is stored in the SQLite database along with the attack type, confidence, severity, threat score and prediction time."

Professor: What happens in Admin Panel?

Answer:

"The Admin Dashboard provides a real-time overview of the system."

Total Requests

Example

150

Meaning

All requests received.

Malicious Requests

Example

38

Meaning

Number of detected attacks.

Safe Requests

Example

112

Meaning

Normal traffic.

Recent Logs

Displays

Time

Request

Attack

Confidence

Severity

Meaning

Administrator can monitor recent activity.

Attack Distribution

Shows

SQL



XSS



Traversal



Meaning

Which attack occurs most frequently.

Global Attack Map

Explain honestly:

"For demonstration purposes, attacks are displayed on a world map using sample geographic locations. In a production deployment, these locations would come from the real client IP address."

Export Logs

Explain

"The administrator can download the attack logs as a CSV file for reporting or further analysis."

Simulate Attack

Explain

"This feature generates sample attack payloads to test the system without requiring real malicious traffic."

Professor: Where is the AI?

Answer confidently:

"The AI is the trained Bi-Directional LSTM model. The dashboard, API, and browser extension do not make predictions themselves—they simply send the request to the model. The model performs the classification, and the application displays the results."

Professor: Why is this Deep Learning?

Answer:

"Unlike rule-based systems that rely on manually written patterns, this project uses a neural network trained on labeled examples. During prediction, the model analyzes the numerical representation of the request and predicts the attack class based on patterns it learned during training."

One-line summary to end the demo

"In short, a request enters the system, it is converted into numerical sequences, analyzed by the Bi-Directional LSTM model, classified into one of five attack categories, assigned a confidence and threat score, logged into the database, and presented to the user and administrator through the dashboard."